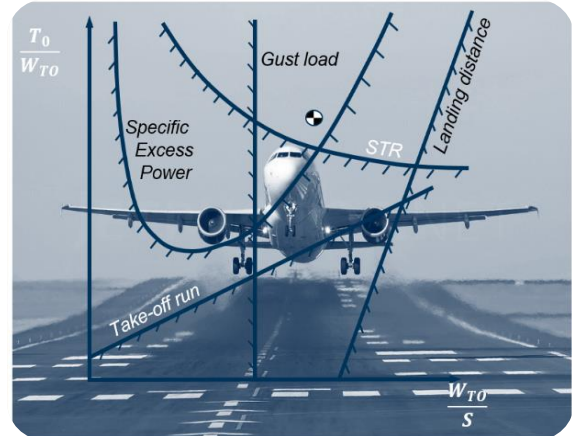


Tutorial VII – Design Chart

By enabling a graphical representation of customer requirements and certification requirements, the design chart becomes an essential tool in aircraft preliminary design. Two key aircraft properties can be determined by the chart: wing loading and thrust-to-weight ratio. This exercise shows the general procedure for creating limit curves on the design chart using a civil transport jet with three engines as an example.



Basics

-  What is the design chart and what do we use it for in aircraft design?
-  Which aircraft parameters are drawn on the x- and y-axis of the design chart?
-  Explain the effect each of these two parameters has on aircraft performance.
-  What typical limit curves can be represented in the design chart?

Calculation tasks

A modern, three-engine passenger jet is to be developed based on the fundamental aircraft design methods. Determine the following limit curves and present them graphically in a T_0/w_{TO} vs. w_{TO}/S chart. Attention: Do not use the A320 dataset for this tutorial!

- 1) Take-off distance (ground roll only) of 1400 m from an airport at mean sea level altitude assuming that $\bar{T} = T_0$.
- 2) Minimum prescribed climb gradient in the case of an engine failure (OEI) for trijets at an altitude of $h = 0$ m at $V = 86$ m/s with a maneuvering weight of $w_{man} = 0.990 \cdot w_{TO}$.
- 3) Service ceiling of $h = 12.5$ km at $M = 0.78$ with $w_{man} = 0.956 \cdot w_{TO}$ (residual climb rate of $SEP = 500$ ft/min necessary).
- 4) Maximum flight Mach number of $M = 0.90$ at an altitude of $h = 10$ km with $w_{man} = 0.850 \cdot w_{TO}$.
- 5) Vertical gust load factor of $\Delta n_z \leq 1.5$ at an altitude of $h = 4000$ m at $V = 230$ kn in the case of a vertical wind gust with velocity $w_g = 15$ m/s at $w_{man} = 0.750 \cdot w_{TO}$.
- 6) 1-g minimum speed of $V_S = 130$ kn at mean sea level with $w_{man} = 0.75 \cdot w_{TO}$.
- 7) Deceleration distance of 650 m at landing with a weight of $w_L = 0.75 \cdot w_{TO}$ at an airport at mean sea level.
- 8) Finally, suggest a design point, which fulfills the above requirements.

Assumptions and Additional Data

Propulsion	Aircraft configuration	General
$\rho_i = 1.225 \text{ kg/m}^3$	$AR = 10$	$g = 9.81 \text{ m/s}$
$n_p = 0.7$	$e_C = 0.85$	Unit conversion
$V_i = 170 \text{ m/s}$	$e_{TO} = 0.80$	$1 \text{ m/s} = 1.944 \text{ kn}$
$n_V = -0.4$		$1 \text{ ft} = 0.3048 \text{ m}$

Task 1: Take-off constraint

- $C = 1.15$
- $C_{L,max,TO} = 2$
- $C_{D0,TO} = 0.02$
- $C_{L,TO} = 1.152$
- $\mu_f = 0.02$

Task 2: OEI-climb constraint

- $C_{D0} = 0.02$
- Apply the lifting line theory for $C_{L\alpha}$
- C_{Di} : Polar break due to high C_L -value
- Excerpt of CS-25 regulations for OEI case second segment climb:

(b) *Take-off; landing gear retracted.* In the take-off configuration existing at the point of the flight path at which the landing gear is fully retracted, and in the configuration used in CS 25.111 but without ground effect,

(1) The steady gradient of climb may not be less than 2.4% for two-engined aeroplanes, 2.7% for three-engined aeroplanes and 3.0% for four-engined aeroplanes, at V_2 with –

(i) The critical engine inoperative, the remaining engines at the take-off power or thrust available at the time the landing gear is fully retracted, determined under CS 25.111, unless there is a more critical power operating condition existing later along the flight path but before the point where the aeroplane reaches a height of 122 m (400 ft) above the take-off surface (see AMC 25.121(b)(1)(i)); and

(ii) The weight equal to the weight existing when the aeroplane's landing gear is fully retracted, determined under CS 25.111.

Task 3: Service ceiling constraint

- $C_{D0} = 0.018$

Task 4: Mach number constraint

- $C_{D0} = 0.019$

Task 5: Gust sensitivity constraint

- Apply the lifting line theory for $C_{L\alpha}$

Task 6: Minimum speed constraint

- $C_{L,max} = 2.4$

Task 7: Landing distance constraint

- $C_{L,max,L} = 2.4$
- $a = 3.5 \text{ m/s}^2$
- No thrust reverser installed

Equations

Take-off acceleration:

$$s_{acc} = \frac{1}{2} \cdot \bar{a} \cdot t^2 = \frac{V_{TO}^2}{2 \cdot \bar{a}}$$

$$\frac{\bar{w}}{g} \cdot \bar{a} = \bar{T} - \bar{D}$$

$$\bar{D} = \bar{D}_f + \bar{D}_{aer}$$

$$\bar{D}_f = \mu_f \cdot (w_{TO} - \bar{L}_{TO})$$

$$D_{aer} = D_{0|clean} + \Delta D_{flap} + \Delta D_{LG} + D_i$$

$$\bar{L}_{TO} = \frac{L_{TO}(V=0) + L_{TO}(V=V_{TO})}{2}$$

$$\bar{q} = \frac{q(V=0) + q(V=V_{TO})}{2}$$

$$C_{Di} = \frac{(C_{L_{TO}} - C_{L,0})^2}{\pi \cdot AR \cdot e} - \frac{C_{L,0}^2}{\pi \cdot AR \cdot e}$$

$$C_{L,0} \approx \frac{1}{3} \cdot C_{L_{TO}}$$

$$V_{TO} = C \cdot V_S$$

Landing deceleration:

$$s_{dec} = \frac{1}{2} \cdot \bar{a} \cdot t^2 = \frac{V_{TD}^2}{2 \cdot \bar{a}}$$

$$V_{ap} = 1.3 \cdot V_S$$

$$V_{TD} = 1.15 \cdot V_S$$

Other equations:

$$\Delta\alpha = \frac{w_g}{V}$$

$$\Delta\bar{a}_{TR} = \frac{T_0 \cdot \eta_{TR}}{w_{TD}} \cdot g$$

$$SEP \approx \gamma \cdot V$$